Music Mood Detection

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***Abstract*—Music mood classification plays a critical role in Music Information Retrieval (MIR), enabling users with personalized recommendations of songs in streaming platforms. With the rise of deep learning, various approaches incorporating lyrics and audio have been on the rise. In this study, we replicate and evaluate the work of Pyrovolakis et al. [1], focusing on their lyric-based and audio-based deep learning models for mood classification across four categories: happy, angry, sad, and relaxed. We used two MoodyLyrics datasets for songs-mood classifications and retrieve lyrics for them via the Lyrics.Ovh API and the Spotify 1 Million Tracks Dataset. We also collected audio previews of songs using the Deezer API. For lyric classification, we implemented models using various embedding techniques (BoW, TF-IDF, Word2Vec, GloVe, and BERT) alongside LSTM, feedforward, and Transformer-based architectures. For audio classification, we extracted six features and trained a CNN model. Our results show that BERT-based lyric models performed best, achieving 62.2% accuracy, while audio-based models underperformed significantly, likely due to limited data and computational constraints. Despite challenges, this project provided us insights into the individual contributions of lyrics and audio to mood classification and highlights future directions for improving deep learning emotion-aware music systems.**

I. Introduction and Motivation

Music is an essential part of life in many cultures and is widely recognized for its ability to influence emotions and moods [2]. With over 100 million tracks available on its platform and a user base of 675 million, Spotify alone highlights the vast scale and significance of music consumption [3]. This underscores the need for continued research in Music Information Retrieval (MIR), particularly in mood prediction—a field that has been actively explored for the past two decades [4]. Automatic music mood detection plays a crucial role in streaming services and recommendation systems, enabling more personalized and emotionally resonant suggestions. By associating mood with music, these systems can gain deeper insights into listener preferences, refining recommendations and enhancing the overall user experience.

In this work, we aim to partially replicate and evaluate components of the approach proposed by Pyrovolakis et al. [1], focusing specifically on deep learning techniques for mood classification using either lyrics or audio signals individually. Rather than implementing the full multimodal fusion model, we concentrate on analyzing the performance of separate single-channel architectures. Using the MoodyLyrics dataset, we assess how different deep learning models contribute to mood prediction accuracy using the audio and lyric features, and identify areas for potential improvement.

II. Related Work

Early approaches to music mood classification primarily relied on handcrafted features and traditional machine learning techniques. In the first half of the 20th century, the work by Hevner [5] was one of the early works exploring correlations between emotions and musical characteristics. Then as time went on, more features based and machine learning methods has started to develop, with features chosen from the lyrics of songs or the audio [4]. For instance, in 2003, the work by Li and Ogihara [6] used audio signal processing features and performed a multi—label classification. Currently, with the advancement of deep learning, neural network-based approaches have also gained prominance in automatic music mood detection and also include fewer feature engineering [4]. This is especially fitting since neural networks excel at capturing complex patterns, making them well-suited for tasks involving auditory data, lyrical analysis, or multimodal approaches that integrate both audio and lyrics.

Transformer-based language models, particularly BERT [7], have greatly advanced natural language processing by enabling deep, context-aware understanding of text. Pretrained using masked token prediction and next sentence prediction, BERT learns rich bidirectional representations that capture complex semantic and syntactic patterns. These strengths make it well-suited for analyzing emotionally nuanced text, such as song lyrics. When fine-tuned for music mood classification, BERT can model subtle emotional cues in lyrics, aiding in accurate mood detection.

OpenL3 is a deep learning model designed to extract high-level audio features from raw audio signals [8], [9]. It uses transfer learning by leveraging pre-trained networks, such as those trained on large-scale datasets like AudioSet, to generate meaningful representations of audio content. These features capture complex patterns in both the spectral and temporal domains, making OpenL3 suitable for tasks like music mood detection. By learning to represent timbral, harmonic, and rhythmic aspects of audio, OpenL3 helps classify music moods by capturing emotional cues in the music, such as tempo, harmony, and dynamics. This enables more accurate mood classification, especially when combined with other features like lyrics.

A recent work by Pyrovolakis et al. [1] demonstrated various deep learning models, involving audio input, lyrical input and multi-modal inputs. They found that their multi-modal approaches outperform single-channel models in music mood detection. They explored this by comparing single-channel and multi-modal deep learning architectures for classifying songs into mood categories. Their study utilized the MoodyLyrics dataset, consisting of 2,000 songs labeled into four mood classes: happy, angry, sad, and relaxed. They examined convolutional neural networks (CNNs) for audio processing, long short-term memory (LSTM) networks for lyric analysis, and combined the two architectures for their multi-modal architectures. This methodological approach highlighted the potential

of cross-modal feature learning, where complementary audio and text representations could jointly capture richer emotional semantics compared to single systems.

III. Methods

A. Data

We discuss the different data sources for the Lyric and Audio Model. For both approaches, we split the full data sets into 80% for training and 20% for testing.

1) Lyrics: For models classifying mood using lyrics, we use the MoodyLyrics and MoodyLyrics4Q datasets for training. [10], [11] Both datasets combined consists of around 3935 unique songs with a mood label for each one and categorizes songs into four mood classes: happy, angry, sad, and relaxed. The MoodyLyrics dataset's mood classification was built using mood tags sourced from Last.fm, a music recommendation and social tagging platform that collects user-generated metadata. The MoodyLyrics4Q dataset was constructed only based on the songs textual lyrics. These datasets can be downloaded here.

Since the MoodyLyrics and MoodyLyrics4Q datasets, do not contain the actual song lyrics, we used the Spotify Million Playlist Dataset and Lyrics.ovh API to extract as much lyrics as possible. [12], [13] This let us gather lyrics for 3007 out of the 3935 songs, which is still greater than the 2000 songs used in [1]. Similar to the Pyrovolakis et al. paper, we split each of the song's lyrics into four line chunks with the same corresponing mood label as the full song, resulting in 26002 data points. Figure 1 shows that there is not much significant class imbalance for Mood in the datasets before or after splitting up the the lyrics into 4 lines.

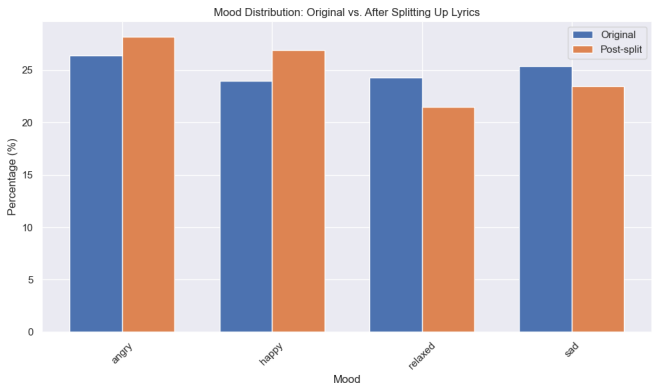


Fig. 1: Class Distribution of the Mood in the Lyrics data set before and after splitting up the lyrics into 4-line chunks.

2) Audio: The MoodyLyrics dataset [10] was used as the foundation for training the classification model based on audio data. To obtain the corresponding audio for the songs in the dataset, we retrieved audio previews using the Deezer API [14], as full-length tracks are copyrighted and not readily accessible. The previews vary in length—some shorter, some longer—but the majority are approximately 30 seconds long. In total, preview audio was successfully collected for 1,887 out of the 2,000 songs in the dataset.

Following the method described by Pyrovolakis et al. [1], each preview was segmented into 10-second clips, with each clip inheriting the mood label of the original song. Only segments with a minimum length of 9 seconds were retained; these were then zero-padded if needed to reach 10 seconds. Any remaining duration shorter than 9 seconds at the end of a preview was discarded (e.g., a 25-second preview yielded two full 10-second clips and discarded the final 5 seconds). All audio files were resampled to 22,050 Hz to ensure consistency during preprocessing and feature extraction.

After segmentation, the dataset consisted of 5663 audio clips. Compared to the data used in the Pyrovolakis et al. paper, which has 37989 10-seconds audio clips, we have significantly less data due to the limited availability of audio. Despite the reduced sample size, the distribution of mood classes across the clips appeared to be relatively uniform, as seen in Fig. 2.

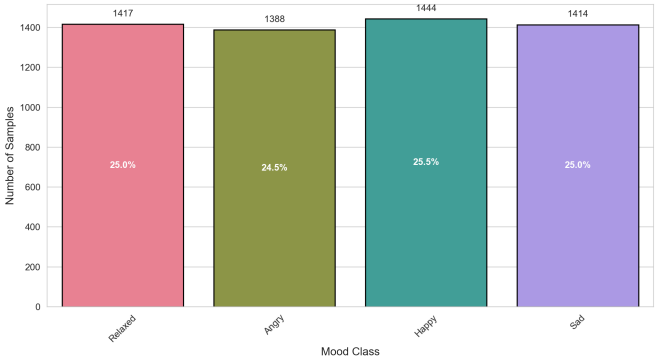


Fig. 2: Class Distribution After Segmenting Audio Files

B. Features

This section explores the different text embeddings employed in the Lyrics models and the different features used in the Audio Model.

1) Lyrics: To extract features from song lyrics, several word embedding techniques were employed. For all of the methods, the lyrics text was preprocessed by being converted to lowercase, removing punctuation and stopwords, and applying stemming to words (eg. extracting "swim" from "swimming").

- **Bag of Words (BoW):** BoW is a text representation method in which a predefined vocabulary is constructed, and each text segment is represented as a vector containing the frequency counts of words from that vocabulary. A vocabulary of the 10,000 most frequent words was constructed. Each lyric was represented as a 10,000-dimensional sparse vector, which was then reduced to a dense vector of size 128 via PCA.
- **Term Frequency–Inverse Document Frequency (TF-IDF):** TF-IDF method reflects how important a word is to a document in a collection, where words with high frequency in document, but low frequency in a collection get higher scores. A vocabulary of the 10,000 most frequent words was used and then there was also a dimension reduction into to a vector of size 128 via PCA.

- **Word2Vec:** Word2Vec is a neural network-based method used to learn continuous vector representations of words, so semantically similar words have similar vectors. Due to a lack of computation power, in our work the implemented Word2Vec model, returns a 128 size vector for each word in the vocabulary and we represent a lyric chunk as the average of all the vector representations of words in that chunk.
- **GloVe (Global Vectors):** This method involves a pretrained word embeddings with a vocabulary of 400,000 words and a 100-dimensional vector representation. [15] Words not present in the vocabulary were assigned zero vectors. Similar to the last method, we represent a lyric chunk as the average of all the vector representations of words in that chunk.
- **BERT Embeddings:** BERT's tokenizer was employed to create contextual vectorizations of the lyrics.

2) Audio: We extract six features from each 10-second audio segment to comprehensively capture the emotional and structural characteristics of music. For each of the 10-second segments, a Short-Time Fourier Transform (STFT) with a 2048-sample window and 512-sample hop length decomposes the audio into time-frequency representations. From here, we calculated the following features:

- **Mel-Spectrogram**: The mel-spectrogram provides a perceptually motivated time-frequency representation by applying a nonlinear mel-scale transformation to the magnitude of the short-time Fourier transform (STFT) of the signal. We use a 60-band representation, and it can mirror the frequency resolution of human hearing. The term Mel refers to a perceptual pitch scale developed through listener-based experiments designed to reflect how humans perceive tonal changes. The name was introduced by Stevens, Volkmann, and Newman [16], and is derived from the word melody. It effectively captures timbral and harmonic nuances essential for mood discrimination.
- **Log-Mel Spectrogram**: We also used 60 bands representation, and it is a logarithmic transformation of Mel-Spectrogram over the frequency axis.
- **Mel-Frequency Cepstral Coefficients (MFCCs)**: For this feature we also use 60 coefficients to match with the Mel-Spectrogram dimension. MFCCs compress spectral information by applying a discrete cosine transform to the mel-spectral energies, resulting in 60 coefficients that compactly represent the spectral envelope. These coefficients are effective in encoding timbral textures that help distinguish different instrumentation and vocal styles.
- **Chroma**: Chroma vectors map spectral content onto the 12 pitch classes of the Western musical scale (C,C#,D,D#,E,F,F#,G,G#,A,A#,B), highlighting harmonic structures like chord progressions and tonality (e.g., major vs. minor keys). To match the dimensionality of other features, the chroma output is upsampled to 60 bins.
- **Tonnetz**: Tonnetz represents the harmonic relationships between pitches in a geometric space. Tonnetz features are derived from chroma features using mathematical transformations that map pitch classes to a six-dimensional space. The six resulting dimensions, associated with perceived emotional valence, are replicated across 60 bins. Tonnetz is useful for capturing tonal qualities like consonance, dissonance, and key changes — all of which can be important cues in mood classification.
- **Spectral Contrast**: Spectral contrast captures the difference between peaks and valleys in six frequency subbands, producing seven values (six local, one global). It reflects the harmonic-to-noise ratio, helping in distinguishing clean, resonant sounds from noisy, dissonant ones. The seven values are upsampled eight times, then padded with zeros, giving the output 60 bins for consistency.

Each feature matrix is time-aligned to 431 frames. These frames are derived from 10-second clips, sampled at 22,050 Hz with 512-sample STFT hops. The matrices are then stacked into a (60, 431, 6) tensor, following the output dimensions from the paper.

C. Model Architecture

Basing from the Pyrovolakis et al. paper, we will reimplement both their lyrical and audio based models. We discuss the different types of Model Architecture below.

1) Lyrics: For lyrics, we implement three neural network architectures: a Recurrent Neural Network with LSTM layers (T_1), a Fully-Connected Neural Network (T_1'), and a Transformer-based model using a pretrained BERT architecture (T_2).

The Fully-Connected model (T_1') is composed of two Fully-Connected Layers. The first layer has 128 neurons with ReLU activation, and the second one has four neurons with softmax activation. No dropout is applied in this model. The LSTM-based model (T_1) begins with a single LSTM layer with 64 units, using tanh as the activation function and sigmoid for the recurrent activation. A dropout rate of 0.2 and a recurrent dropout of 0.2 are applied to the LSTM layer. This is followed by two fully connected layers as in T_1'. The T_1 and T_1' models are shown in Figure‘3.

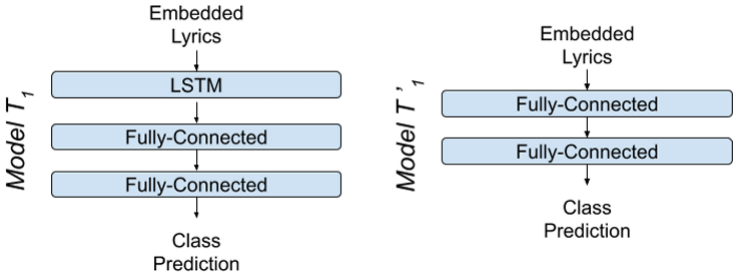


Fig. 3: Architectures of the T_1 and T_1' models, as provided in [1].

Lastly, the Transformer-based model (T_2) is built on the pretrained BERT-base (uncased), which includes 12 Transformer encoder layers, each consisting of multi-head self-attention

(12 heads), layer normalization, and position-wise feedforward layers. On top of BERT, we add two fully connected layer: the first with 256 neurons, sigmoid activation, and a dropout rate of 0.1 and the second with 4 neurons and softmax activation. We trained all models for 20 Epochs, while T_2 was only trained for 2.

2) Audio: For Audio, we implement the Convolutional Neural Network (A_1) as presented in Fig. 4

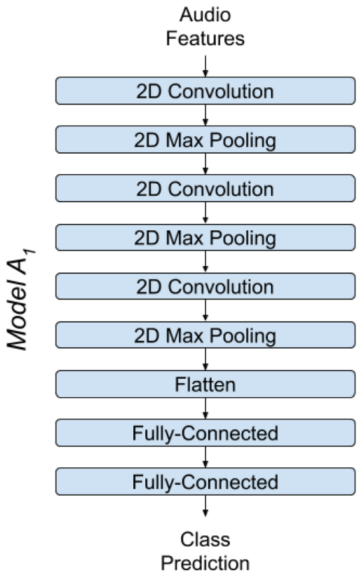


Fig. 4: Architectures of the A_1 model, as provided in [1].

The network consists of three hierarchical convolutional blocks, each comprising a 2D convolutional layer with ReLU activation and L_2 regularization (λ=0.00001) followed by max-pooling. The first layer uses 128 filters and a kernel size of 6x6. Padding is used so that the dimension of the layer's output would be the same as the input's. The second layer employs 256 (5×5) filters with the same padding to identify mid-level features while maintaining resolution, and the final layer uses 512 (4×4) filters to extract high-level representations. All the max-pooling layers use 2x2 window dimensions. The outputs are then flattened and passed through a fully connected layer with strong regularization: a 16-unit ReLU layer with 60% dropout, followed by a softmax output layer for 4-class mood prediction. The model uses Adam optimization (λ=0.001) and runs for 100 epochs.

IV. Results

A. Lyrics

We report the overall Test accuracy of the different Lyric methods, below in Table I and compare it with the percents reported in the Pyrovolakis paper. [1]. We acknowledge that our comparisons with their paper may not be entirely fair, as we used a larger training dataset and, due to computational constraints, adopted a simplified version of the other paper's approach.

To limit overfitting, we selected the number of training epochs such that the training accuracy did not significantly exceed the validation accuracy, in order to reduce overfitting. As a result, Model T_2 was trained for only 2 epochs, since training it for longer led to noticeable overfitting, and the other models were trained for 20 epochs. The training accuracy and validation accuracy is shown in Figure 5.

Model	Embedding Type	Test Accuracy (%)	Pyrovolakis
T_1	GloVe	41.01	53.33
T_1	Word2Vec	41.20	41.66
T_1'	BoW	23.65	65.49
T_1'	TF-IDF	53.41	67.98
T_2	BERT	62.20	69.11

TABLE I: Test accuracy for the different models, compared with evaluation values from Pyrovolakis et al.

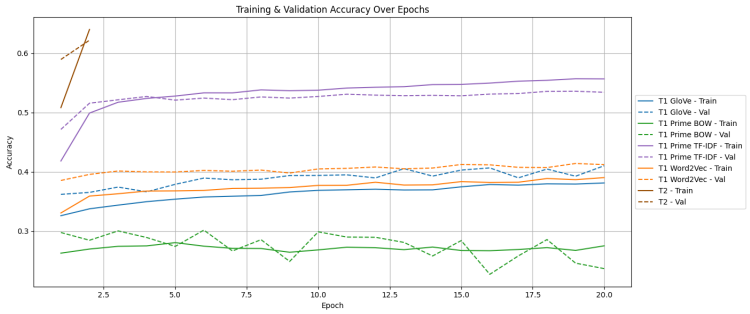


Fig. 5: The Training & Validation Accuracy of all the Lyric Models over Epochs

As shown in Table I, our BERT-based model (T_2) achieved the highest test accuracy among all methods at 62.20%, though it still underperformed compared to the 69.11% reported by Pyrovolakis et al. The T_1 using Word2Vec embeddings model, performed very similar to Pyrovolakis's paper, only less by 0.46%. For the other models, our Test Accuracy was significantly lower than the papers. The largest difference was with the T_1' model using BoW embedding, where we only had 23.65 %, when their paper had 65.49 %. Perhaps this was an error on our part or in interpretation of the model as written in their the paper.

Table II shows how the BERT-based model (T_2) also achieved the lowest test loss at 0.9466, even lower than what was reported by Pyrovolakis et al. Similar result about loss held true for the T_1' model with TF-IDF embeddings. For the other models, our loss values was higher than the papers, supporting the lower accuracy trends.

Model	Embedding Type	Test Loss	Pyrovolakis
T_1	GloVe	1.28	1.064
T_1	Word2Vec	1.26	1.262
T_1'	BoW	2.22	1.287
T_1'	TF-IDF	1.07	1.381
T_2	BERT	0.9466	1.353

TABLE II: Test loss for the different models, compared with evaluation values from Pyrovolakis et al.

B. Audio

We attempted to run the A_1 model with four classes classification and binary classification, using all six features to feed into the model. In the four-class classification task (happy, angry, sad, relaxed), the model achieved only 26% validation accuracy after 100 training epochs (Fig. 6), performing only slightly better than chance level (25%). This performance is significantly lower than the 70% accuracy reported by Pyrovolakis et al. [1] in their original study. This poor performance suggests fundamental limitations in either the model architecture or feature representation's ability to discriminate between the specified mood categories.

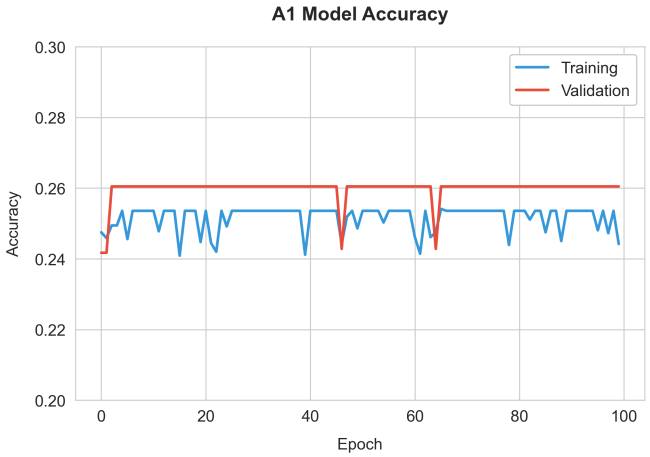


Fig. 6: A_1 model accuracy curve for 4 moods classification

For the binary classification task, moods were grouped into negative (angry, sad) and positive (happy, relaxed) categories, the model's accuracy improved marginally to approximately 50%. However, this result remains equivalent to random chance performance in binary classification, indicating that the model failed to learn meaningful discriminative patterns between the mood classes.

V. Conclusion

Ultimately, our team was able to successfully replicate the models presented in the original paper. Most of the lyrics-based models performed relatively well in comparison to the model in the Pyrovolakis et al. [1] paper, with the exception of the model using Bag-of-Words (BoW) embeddings. The audio-based model did not achieve high accuracy, but this result can be largely attributed to the limited number of available training samples. Despite these challenges, the project gave us valuable hands-on experience in applying deep learning techniques to the task of music mood classification. Music mood detection has practical applications in personalized music recommendations, playlist generation, and emotional wellness platforms. Thus, our project explored how lyrics and audio features contribute to mood perception and how deep learning can model these patterns effectively.

A. Limitations and Future Works

One of the most significant limitations we encountered was the lack of access to full audio data. The songs in the two Moody datasets were copyrighted so we were unable to obtain the complete tracks. We were only able to get the previews of each song. This limitation greatly reduced the number of usable data points, which likely impacted the model's ability to classify mood representations. With access to full-length tracks, we would have been able to generate more segments and train on a bigger dataset, potentially improving the neural network's performance. Similarly, there was also a lack of free online resources to extract lyrics data for all the songs in Moody datasets. If we had more lyrical data, after segmentation of lyrics into 4 lines chunks, this would have led to a significant increase in data points.

Additionally, since the original paper certain design choices made during the feature extraction process may have unintentionally affected model performance. For instance, low-dimensional features such as Tonnetz and Chroma were replicated to match the dimensions of other features, which could introduce redundancy. Furthermore, segments shorter than 9 seconds were discarded, possibly leading to the loss of valuable data. These decisions were made to maintain consistent input dimensions for the model, as the original paper did not clearly specify how these cases were handled. They may have introduced redundancy or led to data loss.

Compute limitations also played a significant role in shaping our approach. Due to limited access to high-performance GPUs, we were unable to train more complex models or experiment with longer training durations, larger batch sizes, or extensive hyperparameter tuning. Consequently, we had to simplify certain aspects of our models. For example, in the Word2Vec and GloVe-based models, we reduced memory usage by averaging word vectors instead of storing the full embedding matrices for each input. For the T_2 model, our use of BERT was constrained by the limited compute credits available on Google Colab, which restricted the number of training runs and prevented us from exploring more extensive augmentation or optimization strategies. Similarly, training the A_1 model posed a challenge, as the input data alone amounted to nearly 4GB—an impractical size for processing on our personal machines. These constraints underscore the importance of having access to more robust computing infrastructure, especially as future work aims to incorporate larger datasets and more sophisticated model architectures.

For future work, we plan to research more and find a way to get the full audio for the songs. We hope that it would help with increasing the traning data points for the model and help improve the accuracy potentially. Once we have the audio based classification model to perform with a decent accuracy, we plan to further explore the lyrics and audio combined model presented in Pyrovolakis's paper. Furthermore, we would try to classify the moods using a Recurrent Neural Network as it is better at processing sequential data compared to Convolutional Neural Network.

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